Universität Bonn

Notes for the lecture

Combinatorics, Graphs, Matroids

held by

Dr. Ulrich Brenner

TEXed by

Daniel Emse

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Corrections and improvements

If you have corrections or improvements, contact me via (`daniel.emse@uni-bonn.de`).

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Lecture

Part I

Combinatorics

Chapter 1

Introduction

1.1 Fundamental Counting Rules

[08.10.2024, Lecture 1]

Goal: Count the number of elements in some finite sets.

Lemma 1.1: Fundamental counting rules

(*i*) *For pairwise disjoint and finite sets* $U_1, \dots, U_n$*, we have*

$$\left| \bigcup_{i=1}^{n} U_i \right| = \sum_{i=1}^{n} |U_i| . \tag{1.1}$$

(*ii*) *For finite sets* $U_1, \dots, U_n$*, we have*

$$|U_1 \times \cdots \times U_n| = \prod_{i=1}^{n} |U_i| . \tag{1.2}$$

(*iii*) *If there is a bijection between two sets* S *and* T*, then* $|S| = |T|$.

Notation. We call a set a n-set if it is a set with cardinality $n \in \mathbb{N}$ (where $0 \in \mathbb{N}$).

Proposition 1.2. *For two* n*-sets* A *and* B*, the number of bijections between* A *and* B *is* $n!$.

Proof. Let U the set of bijections from A to B. Apply induction in n:

$n = 0$: Trivial, because $0! = 1 = |U|$.

$n \mapsto n + 1$: Let $n > 0$ and let $x \in A$. For $i \in B$, let U_i be the set of all bijections $f : A \to B$ with $f(x) = i$. By induction, $|U_i| = (n-1)!$ for each $i \in B$. Then, $U = \bigcup_{i \in B} U_i$, so we get

$$|U| = \sum_{i \in B} |U_i| = n \cdot (n-1)! = n!. \tag{1.3}$$

□

Corollary 1.3. *The number of permutations of an* n*-set is* $n!$. □

Proposition 1.4. *The number of mappings from a* k*-set* A *to an* n*-set* B *is* n^k.

Proof. There is a bijection between the set of all mappings from A to B and the set $\underbrace{B \times \cdots \times B}_{k \text{ times}}$.

The number of these mappings is by Lemma 1.1.(*ii*) just $|B|^k = n^k$. □

Notation. We denote the set of all mappings from A to B by B^A.

Proposition 1.5. *The number of subsets of an n-set is 2^n.*

Proof. For an n-set A, there is a bijection between its power set and the mappings from A to $\{0,1\}$. Thus, the power set has 2^n elements. □

Notation.

(*i*) We denote the power set of a set A by 2^A.

(*ii*) The *binomial coefficient* $\binom{n}{k}$ is defined as the number of k-subsets of a n-set (for $k, n \in \mathbb{N}$).

What does *counting* mean? $\rightsquigarrow$ Possible answers:

(*i*) a direct, closed formula,

(*ii*) a sum,

(*iii*) a recursive formula

Example. Consider the number y_n of $0-1-2$-strings of length n with an even number of 1s and an odd number of 2s. Answers of types (*ii*) and (*iii*):

- We have
$$y_n = \sum_{i=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{2}{2i+1} 2^{2i}. \tag{1.4}$$
- We have
$$y_n = 3^{n-1} - y_{n-1} \quad (n > 0). \tag{1.5}$$

The proof of both formulas is exercise 1.2.

1.2 Elementary Counting Coefficients

Proposition 1.6. *For $n, k \in \mathbb{N}$ with $k \le n$, we have*

(*a*) $\binom{n}{k} = \binom{n}{n-k}$

(*b*) $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ *for* $k \ge 1$

(*c*) $\binom{n}{k} = \frac{n!}{(n-k)!k!}$.

Consider three kinds of rows in Pascal's triangle:

Row sums $\sum_{k=0}^{n} \binom{n}{k} = 2^n$ (both terms count the number of subsets of an n-set)

Column sums $\sum_{m=0}^{n} \binom{m}{k} = \binom{n+1}{k+1}$ for $n, k \in \mathbb{N}$ (where $\binom{m}{k}$ is the number of all $(k+1)$-subsets of $\{1, \ldots, n+1\}$ with the property that $m+1$ is the largest element in the subset)

Diagonal sums $\sum_{k=0}^{n} \binom{m+k}{k} = \binom{m+n+1}{n}$ for $m, n \in \mathbb{N}$ (for $k \in \{0, \ldots, n\}$, where $\binom{m+n}{k}$ is the number of n-subsets of $\{1, \ldots, m+n+1\}$ where $m+k+1$ is the largest number that is **not** contained in the subset)

Proposition 1.7. *For $x, y \in \mathbb{R}$, we have $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$.*

Proof. The coefficients of $x^k y^{n-k}$ in $(x+y)^n$ is $\binom{n}{k}$. □

Definition 1.8: $S_{n,k}$: Stirling number of the second kind

For $n, k \in \mathbb{N}$ let $S_{n,k}$ be the number of ways to partition an n-set in to k non-empty sets. The number $S_{n,k}$ is called *Stirling number of the second kind.*

Observation. For $n > 0$: $S_{n,0} = 0, S_{n,1} = n, S_{n,2} = 2^{n-1} - 1$.

Proposition 1.9. *For $n, k \in \mathbb{N} \setminus \{0\}$ we have $S_{n,k} = S_{n-1,k-1} + k \cdot S_{n-1,k}$.*

Proof. Let M be an n-set and $x \in M$. Then, the set of partitions of n into k subsets can be decomposed in the set of partitions with the set $\{x\}$ (there are $S_{n-1,k-1}$ of them) and the set of partitions where x is an element in a set with more than one element (there are $k \cdot S_{n-1,k}$ of them) □

[08.10.2024, Lecture 1]
[10.10.2024, Lecture 2]

Next goal: Count the number of mappings from an n-set to an r-set under certain constraints. To this end, we collect *counting coefficients*. We know already:

- $\binom{n}{k}$
- $S_{n,k}$: Number of ways to partition an n-set into k non-empty sets

Definition 1.10: $P_{n,k}$: Number of ways writing n as sum of k integers

For $n, k \in \mathbb{N}$, let $P_{n,k}$ be the number of ways to write n as the sum of k positive integers (without considering the order of the summands).

Example. $P_{6,3} = 3$, because the only possible partitions are: $6 = 4+1+1 = 3+2+1 = 2+2+2$.

Observation. For $n > 0$, we have $P_{n,0} = 0, P_{n,1} = 1, P_{n,2} = \lfloor \frac{n}{2} \rfloor$.

We can also consider *ordered partitions of numbers*:

Example. There are 6 ordered 3-partitions of the number 5: $5 = 3+1+1 = 1+3+1 = 1+1+3 = 2+2+1 = 2+1+2 = 1+2+2$.

Proposition 1.11. *The number of ordered partitions of a positive integer n into k summands is $\binom{n-1}{k-1}$.*

Proof. Write n as $n = \underbrace{1 + \cdots + 1}_{n \text{ times}}$. Any choice of $k-1$ of the $n-1$ „+“-signs gives you an ordered partition of n into k summands. Tis is a one-to-one correspondance. □

1.3 Assignments

Observations.

(*i*) The number of *surjective mappings* from an n-set N to an r-set R is $r! \cdot S_{n,r}$.

Proof. There are $S_{n,r}$ ways to partition n into r preimages and for each such partiton, we get $r!$ ways to assign the preimages to the elements of R. □

(*ii*) The number of *injective mappings* from an n-set N to an r-set R is

$$r(r-1)(r-2)\cdots(r-n+1). \tag{1.6}$$

Notation. For $n, r \in \mathbb{N}$, we call

$$r^{\underline{n}} := r \cdot (r-1) \cdot (r-2) \cdots (r-n+1) \tag{1.7}$$

the *n-th falling factorial* of r.

Analogously, the *n-th rising factorial* of r is

$$r^{\overline{n}} := r \cdot (r+1) \cdot (r+2) \cdots (r+n-1). \tag{1.8}$$

Definition 1.12: Multiset

A *k-multiset* over an n-set A is a set $B \subseteq A \times \mathbb{N}$ such that $\sum_{(a,i)\in B} i = k$ and $|\{(a,i) \in B \mid i \in \mathbb{N}\}| = 1$ for all $a \in A$.

$\rightsquigarrow$ Multisets can be considered as a set that may contain an element more than once. We write again $\{a_1, \ldots, a_k\}$ for a k-multiset (so $a_i = a_j$ is possible for $i \neq j$).

Proposition 1.13. *The number of k-multisets over an n-set is $\frac{n^{\overline{k}}}{k} = \binom{n+k-1}{k}$.*

Proof. There is a bijection f between the set of all k-multisets over $\{1, \ldots, n\}$ and the set of all k-subsets of $\{1, \ldots, n+k-1\}$. For a k-multiset $\{a_1, \ldots, a_k\}$ with $a_1 \leq a_2 \leq \cdots \leq a_{k-1} \leq a_k$, set $f(\{a_1, \ldots, a_j\}) = \{a_1, a_2+1, a_3+2, \ldots, a_k+(k-1)\}$. It is easy to see that this defines a bijection. □

Proposition 1.14. *For $n, r \in \mathbb{N}$, we have $r^n = \sum_{k=0}^{n} S_{n,k} r^{\underline{k}}$.*

Proof. For two sets A and B, let $\mathrm{Surj}(A,B)$ be the set of all surjective mappings from A to B. We know from Observation (i) that $|\mathrm{Surj}(A,B)| = |B|! \cdot S_{|A|,|B|}$. Let N be an n-set and R an r-set. Then, r^n is the number of mappings from N to R. We can partition

$$\begin{aligned} r^n &= \sum_{A\subseteq R} |\mathrm{Surj}(N,A)| = \sum_{k=0}^{r} \sum_{\substack{A\subseteq R \\ |A|=k}} |\mathrm{Surj}(N,A)| \\ &= \sum_{k=0}^{r} \binom{n}{k} k! S_{n,k} = \sum_{k=0}^{r} S_{n,k} r^{\underline{k}} = \sum_{k=0}^{n} S_{n,k} r^{\underline{k}} \end{aligned} \tag{1.9}$$

where the last step is possible because $S_{n,k} = 0$ for $k > n$ and $r^{\underline{k}} = 0$ for $k > r$. □

elements distinguishalbe		Mapping $f : N \to R$			
N	R	arbitrary	injective	surjective	bijective
Yes	Yes	r^n	$r^{\underline{n}}$	$r! S_{n,r}$	$\begin{matrix} 0 & (n\neq r) \\ n! & (n=r) \end{matrix}$
No	Yes	$\frac{r^{\overline{n}}}{n!}$	$\binom{r}{n}$	$\binom{n-1}{r-1}$	$\begin{matrix} 0 & (n\neq r) \\ 1 & (n=r) \end{matrix}$
Yes	No	$\sum_{k=1}^{r} S_{n,k}$	$\begin{matrix} 0 & (r<n) \\ 1 & (r\geq n) \end{matrix}$	$S_{n,r}$	$\begin{matrix} 0 & (n\neq r) \\ 1 & (n=r) \end{matrix}$
Yes	Yes	$\sum_{k=1}^{r} P_{n,k}$	$\begin{matrix} 0 & (r<n) \\ 1 & (r\geq n) \end{matrix}$	$P_{n,r}$	$\begin{matrix} 0 & (n\neq r) \\ 1 & (n=r) \end{matrix}$

Table 1.1: Number of mappings from an n-set N to an r-set R with and without additional constraints

First row completely covered.

Second row
- $\frac{r^{\overline{n}}}{n!}$ is the number of r-multisets of an n-set.
- $\binom{n-1}{r-1}$ is the number of ordered partitions of n into r summands.

1.4 Generalized Counting Coefficients

Definition 1.15: Generalized factorials and coefficients

For $r \in \mathbb{C}$ and $k \in \mathbb{N}$, we define

(*i*) $r^{\underline{n}} := r \cdot (r-1) \cdot (r-2) \cdots (r-k+1)$ (with $r^{\underline{0}} = 1$)

(*ii*) $r^{\overline{n}} := r \cdot (r+1) \cdot (r+2) \cdots (r+k-1)$ (with $r^{\overline{0}} = 1$)

For $r \in \mathbb{C}, k \in \mathbb{Z}$, we define

$$\binom{r}{k} = \begin{cases} \frac{r^{\underline{k}}}{k!} & k \geq 0 \\ 0 & k < 0. \end{cases} \tag{1.10}$$

Proposition 1.16. *For $r \in \mathbb{C}, k \in \mathbb{Z}$, we have*

$$\binom{r}{k} = \binom{r-1}{k-1} + \binom{r-1}{k}. \tag{1.11}$$

Proof. If $k \leq 0$: trivial.

For $k > 0$, both sides of the equation are (for a fixed k) polynomials in r of degree k. They are equal for any $r \in \mathbb{N}$, so for an infinite number of values of r. $\rightsquigarrow$ Both polynomials are identical. □

[10.10.2024, Lecture 2]
[15.10.2024, Lecture 3]

Proposition 1.17 (Vandermonde identity)**.** *For $x, y \in \mathbb{C}$ and $n \in \mathbb{N}$, we have*

$$\binom{x+y}{n} = \sum_{k=0}^{n} \binom{x}{k} \binom{y}{n-k}. \tag{1.12}$$

Proof. Again, we can use the polynomial method: For $x, y \in \mathbb{N}$, this is true because both terms describe the number of k-subsets of an $(x+y)$-set. For general values of x and y, the statement follows because both terms are polynomials of degree n and are identical on an infinite number of points. □

Proposition 1.18. *For $r \in \mathbb{C}$ and $k \in \mathbb{Z}$, we have*

$$\binom{-r}{k} = (-1)^k \binom{r+k-1}{k}. \tag{1.13}$$

Proof. We have

$$(-r)^{\underline{k}} = (-r)(-r-1) \cdots (-r-k+1) = (-1)^k r(r+1) \cdots (r+k-1) = (-1)^k r^{\overline{k}}. \tag{1.14}$$

Dividing this equation by $k!$ proves the claim. □

Remark. Together with the equation $\sum_{k=0}^{n} \binom{m+k}{k} = \binom{m+n+1}{n}$ (see the observation about diagonal sums on Pascal's triangle) that is valid for any $m \in \mathbb{C}$, we get

$$\sum_{k=0}^{m} (-1)^k \binom{n}{k} = \sum_{k=0}^{m} \binom{k-n-1}{k} = \binom{m-n}{m} = (-1)^m \binom{n-1}{m}. \tag{1.15}$$

Hence, we have a formula for the alternating sum of a row in Pascal's triangle. Moreover, the equation has a combinatorical interpretation: Both sides count the number of m-subsets of $\{1,\dots,n-1\}$ if m is even (or minus this number if m is odd).

1.5 Permutations

An *n-permutation* is a bijection $\pi : \{1,\dots,n\} \to \{1,\dots,n\}$.

For a permutation π, a *cycle* is a vector $(i_1, i_2, \dots, i_t)$ such that $\pi(i_j) = i_{j+1}$ for $j \in \{1,\dots,t-1\}$ and $\pi(i_t) = i_1$. Hence, a fixed point correspondents to a cycle of length 1.

Definition 1.19: $s_{n,k}$: Stirling number of the first kind

For $n, k \in \mathbb{N}$ let $s_{n,k}$ be the number of n-permutations with k cycles (where we set $s_{0,0} := 1$). The number $s_{n,k}$ is called *Stirling number of the first kind*.

Obviously, for $n > 0$ we have: $s_{n,0} = 0, s_{n,1} = (n-1)!$. As for the Stirling numbers of the second kind, we get a recursion formula.

Proposition 1.20. *For $n, k \in \mathbb{N} \setminus \{0\}$, we have*

$$s_{n,k} = s_{n-1,k-1} + (n-1)s_{n-1,k}. \tag{1.16}$$

Proof. There are $s_{n-1,k-1}$ n-permutations with k cycles where n is a fixed point and $(n-1)s_{n-1,k}$ n-permutatios with k cycles where n is not a fixed point. □

For the Stirling numbers of the second kind, also the number $S_{n,2}$ (for $n > 1$) was easy to compute. To do this for the Stirling numbers of the first kind, we need the harmonic numbers.

Definition 1.21: Harmonic numbers

For $n \in \mathbb{N}$, we define $H_n := \sum_{i=1}^{n} \frac{1}{i}$. We call H_n the *n-th harmonic number*.

It is a well-know fact (that can easily be proved by using the fact that the derivative of $\ln(x)$ is $\frac{1}{x}$) that $H_n \in \Theta(\ln(n))$; more precisely, we have for positive integers n:

$$\ln(n) + \frac{1}{n} \le H_n \le \ln(n) + 1. \tag{1.17}$$

Proposition 1.22. *For $n \in \mathbb{N} \setminus \{0\}$, we have $s_{n,2} = (n-1)!H_{n-1}$.*

Proof. Induction in n. For $n = 1$, we have 0 on both sides. So let $n > 1$. Then:

$$\frac{s_{n,2}}{(n-1)!} = \frac{s_{n-1,1}}{(n-1)!} + \frac{(n-1)s_{n-1,2}}{(n-1)!} = \frac{(n-2)!}{(n-1)!} + \frac{s_{n-1,2}}{(n-2)!} = \frac{1}{n-1} + H_{n-2} = H_{n-1}. \tag{1.18}$$

□

There is a surprising connection between the two kinds of Stirling numbers: They are coefficients that allow to switch between the bases $\{r^0, r^1, r^2, \dots\}$ and $\{r^{\underline{0}}, r^{\underline{1}}, r^{\underline{2}}, \dots\}$ of the vector space of the polynomials. We know already the following equation for $n, r \in \mathbb{N}$:

$$r^n = \sum_{k=0}^{n} S_{n,k} r^{\underline{k}}. \tag{1.19}$$

By the polynomial method, we can generalize this result to $r \in \mathbb{C}$. For the other direction (i.e. for writing $r^{\underline{n}}$ as a linear function of the polynomials r^k), we can use the Stirling numbers of the first kind:

Proposition 1.23. *For $n \in \mathbb{N}$ and $r \in \mathbb{C}$, we have*

$$r^{\underline{n}} = \sum_{k=0}^{n} (-1)^{n-k} s_{n,k} r^k. \tag{1.20}$$

Proof. Induction in n. For $n = 0$, both sides of the equation are equal to 1. Hence, assume $n > 0$. We generalize the definition of $s_{n,k}$ to negative values of k by setting $s_{n,k} := 0$ for $k < 0$ and get:

$$\begin{aligned} r^{\underline{n}} &= (r-n+1) r^{\underline{n-1}} = (r-n+1) \sum_{k=0}^{n-1} (-1)^{n-1-k} s_{n-1,k} r^k \\ &= \sum_{k=0}^{n-1} (-1)^{n-1-k} s_{n-1,k} r^{k+1} + \sum_{k=0}^{n-1} (-1)^{n-k} (n-1) s_{n-1,k} r^k \\ &= \sum_{k=1}^{n} (-1)^{n-k} s_{n-1,k-1} r^k + \sum_{k=0}^{n} (-1)^{n-k} (n-1) s_{n-1,k} r^k \\ &= \sum_{k=0}^{n} (-1)^{n-k} (s_{n-1,k-1} r^k + (n-1) s_{n-1,k} r^k) = \sum_{k=0}^{n} (-1)^{n-k} s_{n,k} r^k \end{aligned} \tag{1.21}$$

□

We categorize permutations not only be the number of cycles but also on the number of cycles of a certain length.

Notation. For a permutation π, there let $b_i(\pi)$ be the number of cycles of length i (for $i \in \{1, \dots, n\}$), and let $b(\pi) := \sum_{i=1}^{n} b_i(\pi)$ be the total number of cycles. The *type of a permutation* π is the formal term $t(\pi) = 1^{b_1(\pi)} 2^{b_2(\pi)} \cdots n^{b_n(\pi)}$ where we skip the numbers i with $b_i(\pi) = 0$.

Observation. $\sum_{i=1}^{n} i b_i(\pi) = n$.

Proposition 1.24. *There are $\sum_{k=0}^{n} P_{n,k}$ types of n-permutations.*

Proof. For every type $1^{b_1(\pi)} 2^{b_2(\pi)} \cdots n^{b_n(\pi)}$, we find a representation of n as a sum of $b(\pi)$ positive integers where the number i occurs $b_i(\pi)$ times as a summand. This leads to a bijection between the set of types of an n-permutations and the set of partitions of n. □

Proposition 1.25. *There are*

$$\frac{n!}{b_1! \cdots b_n! 1^{b_1} 2^{b_2} \cdots n^{b_n}} \tag{1.22}$$

n-permutations of type $1^{b_1} 2^{b_2} \cdots n^{b_n}$.

Proof. First consider the empty cycles

$$\underbrace{(\cdot)(\cdot)\dots(\cdot)}_{b_1 \text{ times}} \underbrace{(\cdot\cdot)(\cdot\cdot)\dots(\cdot\cdot)}_{b_2 \text{ times}} \underbrace{(\cdots)(\cdots)\dots(\cdots)}_{b_3 \text{ times}} \dots \tag{1.23}$$

There are $n!$ ways to fill these cycles with numbers from $\{1, \dots, n\}$ (without repeating numbers). Moreover, each such assignment yields an n-permutation of type $1^{b_1} 2^{b_2} \cdots n^{b_n}$. However, for each $i \in \{1, \dots, n\}$, all assignments that differ just in the order of the b_i cycles lead to the same permutation. And for each cycle of length i, there are i assignments giving the same permutation

because there are i ways to choose the first element in the cycle. Hence, for each n-permutation π, there $b_1! \cdots b_n! 1^{b_1} 2^{b_2} \cdots n^{b_n}$ assignments that encode π. □

In particular, we have

$$s_{n,k} = \sum_{(b_1,\ldots,b_n)} \frac{n!}{b_1! \cdots b_n! 1^{b_1} 2^{b_2} \cdots n^{b_n}} \tag{1.24}$$

where we sum over all vectors $(b_1, \ldots, b_n)$ with $\sum_{i=1}^n i b_i = n$ and $\sum_{i=1}^n b_i = k$.

Similarly:

$$n! = \sum_{(b_1,\ldots,b_n)} \frac{n!}{b_1! \cdots b_n! 1^{b_1} 2^{b_2} \cdots n^{b_n}} \tag{1.25}$$

where we sum over all vectors $(b_1, \ldots, b_n)$ with $\sum_{i=1}^n i b_i = n$.

[15.10.2024, Lecture 3]
[17.10.2024, Lecture 4]

1.6 Further counting principles

1.6.1 Inclusion-exclusion principle

Proposition 1.26. *Let $A_1, \ldots, A_n$ be finite sets. Then,*

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{r=1}^n (-1)^{r-1} \sum_{1 \le i_1 < \cdots < i_r \le n} \left| \bigcap_{j=1}^r A_{i_j} \right|. \tag{1.26}$$

Proof. For each $a \in \bigcup_{i=1}^n A_i$, we have to show that a is counted on the right-hand side of the equation exactly once. Let $k = |\{i \in \{1, \ldots, n\} \mid a \in A_i\}|$, so a is counted $\binom{k}{r}$ times in the sum

$$\sum_{1 \le i_1 < \cdots < i_r \le n} \left| \bigcap_{j=1}^r A_{i_j} \right| \tag{1.27}$$

for $r \in \{1, \ldots, n\}$, so in total, a is counted $\sum_{r=1}^k (-1)^{r-1} \binom{k}{r}$ times. With that, we get

$$0 = (-1+1)^k = \sum_{r=0}^k \binom{k}{r} (-1)^r 1^{k-r} = 1 - \sum_{r=1}^k (-1)^{r-1} \binom{k}{r}, \tag{1.28}$$

so a is counted $\sum_{r=1}^k (-1)^{r-1} \binom{k}{r} = 1$ times. □

Proposition 1.27. *The number of fixed-point-free n-permutations is*

$$D_n := n! \sum_{r=0}^n \frac{(-1)^r}{r!}. \tag{1.29}$$

Proof. Let A_i be the set of al n-permutations π with $\pi(i) = i$. The number of fixed-point-free n-permutations is

$$n! - \left| \bigcup_{i=1}^n A_i \right| = n! - \sum_{r=1}^n (-1)^{r-1} \sum_{1 \le i_1 < \cdots < i_r \le n} \left| \bigcap_{j=1}^r A_{i_j} \right|. \tag{1.30}$$

For $1 \le i_1 < \cdots < i_r \le n$, we have $\left| \bigcap_{j=1}^r A_{i_j} \right| = (n-r)!$. There are $\binom{n}{r}$ ways to choose $i_1, \ldots, i_r$ with $1 \le i_1 < \cdots < i_r \le n$ (for $r \in \{1, \ldots, n\}$). Hence, the number of fixed-point-free

n-permutations is

$$\begin{aligned} n! - \sum_{r=1}^{n} \binom{n}{r} (-1)^r (n-r)! &= \sum_{r=0}^{n} \binom{n}{r} (-1)^r (n-r)! \\ = \sum_{r=0}^{n} \frac{n!}{r!(n-r)!} (-1)^r (n-r)! &= n! \sum_{r=0}^{n} \frac{(-1)^r}{r!}. \end{aligned} \tag{1.31}$$

□

The number D_n is called *derangement number*.

Remark. We have

$$\lim_{n\to\infty} \sum_{r=1}^{n} \frac{(-1)^r}{r!} = \frac{1}{e}, \tag{1.32}$$

so for large n, the fraction of the fixed-point-free n-permutations of all n-permutations is approximately $\frac{1}{e}$.

Proposition 1.28. *For $n \in \mathbb{N}$, the number of ways to write n as the sum of odd, natural numbers is the number of ways to write n as a sum of different, positive integers.*

Proof. For $n \in \mathbb{N} \setminus \{0\}$, let

$$p(n) := \sum_{k=1}^{n} P_{n,k} \tag{1.33}$$

be the number of ways to write n as any sum of positive integers. Hence, the number of partitions into odd summands and the number of partitions into different summands is

$$p(n) - \sum_{r=1}^{n} (-1)^{r-1} \sum_{1 \le i_1 < \cdots < i_r \le n} p\left(n - \sum_{j=1}^{r} 2i_j\right) : \tag{1.34}$$

Let A_i for the odd summands be the number of partitions into summands where $2i$ is one of those, so $|A_i| = p(n-2i)$; and let A_i be for the different summands be the number of partitions into summands where i occurs twice, so $|A_i| = p(n-2i)$.[1] □

1.6.2 Principle of doubly counting

For a relation $R \subseteq S \times T$ where S and T are finite, we can count $|R|$ in two different ways:

$$\sum_{s\in S} |\{t \in T \mid (s,t) \in R\}| = \sum_{t\in T} |\{s \in S \mid (s,t) \in R\}|. \tag{1.35}$$

Example.

(i) Count the average number of divisiors of the integers in $\{1,\dots,n\}$, i.e.

$$\frac{1}{n} \sum_{s\in\{1,\dots,n\}} |\{t \in \{1,\dots,n\} \mid \exists k \in \mathbb{N} : s = kt\}|, \tag{1.36}$$

so we set $S = T = \{1,\dots,n\}$ and define $R \subseteq S \times T$ by $(s,t) \in R$ iff $t|s$. From that, we get that the average number of divisiors is

$$\frac{1}{n} \sum_{t\in T} |\{s \in S \mid (s,t) \in R\}| = \frac{1}{n} \sum_{t\in T} \left\lfloor \frac{n}{t} \right\rfloor \in \left[\sum_{i=1}^{n} \frac{1}{i} - 1, \sum_{i=1}^{n} \frac{1}{n} \right] \approx H_n \approx \ln(n). \tag{1.37}$$

[1] Student: „Is this useful in any way?“ – Brenner: **laughs** „In the combinatorics part, this is always a difficult question. [...] So yes, for combinatorics, it *can* be useful. But it is an interesting result.“ – 17.10.2024

(*ii*) Claim: A simple, undirected graph G that does not contain a cycle of length 4 has at most $\lfloor \frac{n}{4}\left(n+\sqrt{4n-3}\right) \rfloor$ edges (where $n = |V(G)|$ is the number of nodes).

Proof. For $S := V(G)$ and $T := \{\{v,w\} \subseteq V(G) \mid v \neq w\}$, we define $R \subseteq S \times T$ by $(u,\{v,w\}) \in R$ iff $\{u,v\} \in E(G)$ and $\{u,w\} \in E(G)$. Doubly counting yields

$$\sum_{s\in S} |\{t \in R \mid (s,t) \in R\}| = \sum_{u \in V(G)} \binom{|\delta_G(u)|}{2} = \sum_{t \in T} |\{s \in S \mid (s,t) \in R\}| \leq |T| = \binom{n}{2},$$

where $\delta_G(u)$ is the set of neighbours of u. With this, we get

$$\sum_{u\in V(G)} |\delta_G(u)|^2 \leq n(n-1) + \sum_{u \in V(G)} |\delta_G(u)|\,. \tag{1.38}$$

Set $\alpha(u) := \frac{2m}{n} - |\delta_G(u)|$ for $u \in V(G), m = |E(G)|$. Then, we get

$$\begin{aligned} \sum_{u\in V(G)} |\delta_G(u)|^2 &= \sum_{u\in V(G)} \left(\frac{2m}{n} - \alpha(u)\right)^2 \\ &= \sum_{u\in V(G)} \left(\frac{2m}{n}\right)^2 - \frac{4m}{n} \underbrace{\sum_{u\in V(G)} \alpha(u)}_{=0} + \underbrace{\sum_{u\in V(G)} \alpha(u)^2}_{\geq 0} \\ &\geq \frac{4m^2}{n} \Rightarrow \frac{4m^2}{n} \leq n(n-1) + 2m \Rightarrow m \leq \left\lfloor n + \sqrt{4n-3} \right\rfloor. \end{aligned} \tag{1.39}$$

□

1.6.3 Pigeonhole principle

For an n-set N and an r-set R with $n > r$ and a mapping $f : N \to R$, there is an $a \in R$ with $|f^{-1}(n)| \geq \left\lfloor \frac{n-1}{r} \right\rfloor + 1 \geq 2$.

Example. For natural numbers $a_1, \ldots, a_n$ (with $n \geq 1$), there are $k, l \in \{1, \ldots, n\}$ such that $\sum_{i=k}^{l} a_i$ is an integral multiple of n.

Proof. Let $N = \left\{ \sum_{i=1}^{l} a_i \;\middle|\; l \in \{0, \ldots, n\} \right\}$ and $R = \{0, \ldots, n-1\}$. Definie $f : N \to R$ where $f(n)$ is the remainder of n for division by n. With that, we get $|N| = n+1 > n = |R|$, so there exist $m, l \in \{0, \ldots, n\}$ with $m < l$ and $f\left(\sum_{i=1}^{m} a_i\right) = f\left(\sum_{i=1}^{l} a_i\right)$, so we get $\sum_{i=m+n}^{l} a_i$ is an integral multiple of n. □

Theorem 1.29: Ramsey

For $k, l \in \mathbb{N} \setminus \{0\}$, there is a smallest number $R(k,l)$ such that any graph with at least $R(k,l)$ nodes contains a clique of size k (so a K_k) or a stable set of size l (so the complement contains a K_l).

Proof. Induction in $k + l$. If $k = 1$ or $l = 1$, then $R(k,l) \leq \max\{k,l\}$.

Let $l > 1$ and $k > l$. *Claim:*

$$R(k,l) \leq R(k-1,l) + R(k,l-1). \tag{1.40}$$

Proof of the claim: Let G be a graph with $R(k-1,l)+R(k,l-1)$ nodes. Let $v \in V(G)$ be a node, X the set of neighbours of v and $Y = V(G) \setminus (\{v\} \cup X)$. We have $|X| \geq R(k-1,l)$ or $|Y| \geq R(k,l-1)$.

Case 1: $|X| \geq R(k-1,l)$. If X contains a stable set of size l, we are done. Otherwise, by induction, X contains a clique of size $k-1$. Together with v, this gives us a clique of size k.

Case 2: $|Y| \geq R(k,l-1)$. If Y contains a clique of size k, we are done. Otherwise, by induction, Y contains a stable set of size $l-1$. Together with v, this gives us a stable set of size l. □

The numbers $R(k,l)$ are called *Ramsey numbers*.

[17.10.2024, Lecture 4]
[22.10.2024, Lecture 5]

Observations.

(*i*) $R(k,1) = R(1,l) = 1$ for $k,l \in \mathbb{N} \setminus \{0\}$.

(*ii*) $R(2,m) = R(m,2) = m$ for $m \in \mathbb{N} \setminus \{0\}$.

(*iii*) $R(k,l) \leq R(k-1,l) + R(k,l-1)$ for $k,l \in \mathbb{N} \setminus \{0,1\}$ (see in the proof above).

Theorem 1.30: Upper bound of $R(k,l)$

For $k,l \in \mathbb{N} \setminus \{0\}$, we have

$$R(k,l) \leq \binom{k+l-2}{k-1}. \tag{1.41}$$

Proof. By induction. If $l = 1$, we have $R(k,1) \leq \binom{k-1}{k-1} = 1$ and if $k = 1$, we have $R(1,l) \leq \binom{l-1}{0} = 1$.

For $k,l > 1$, we get

$$R(k,l) \leq R(k-1,l) + R(k,l-1) \leq \binom{k+l-3}{k-2} + \binom{k+l-3}{k-1} = \binom{k+l-2}{k-1} \tag{1.42}$$

by (1.40). □

The exact numbers $R(k,l)$ are hard to compute in general.

Example. We have $R(3,3) = 6$: By Theorem 1.30, we have $R(3,3) \leq \binom{4}{2} = 6$, and we find a counterexample with 5 nodes: A cycle of length 5 (called C_5) is an example which shows that $R(3,3) > 5$. By that, we get $R(3,3) = 6$.

Chapter 2

Computation of sums

2.1 Direct methods

(*i*) *Index transformation:* Sums can be computed in different ways, e.g.

$$\sum_{k=m}^{n} a_k = \sum_{k=m+i}^{n+i} a_{k-i} = \sum_{k=m-i}^{n-i} a_{n+k} = \sum_{k=0}^{n-m} a_{m+k} = \sum_{k=0}^{n-m} a_{n-k}. \tag{2.1}$$

Example. Compute $S_n := \sum_{k=0}^{n} ka = \sum_{k=0}^{n}(n-k)a$, so we get

$$\Rightarrow 2S_n = \sum_{k=0}^{n} ka + (n-k)a = \sum_{k=0}^{n} na = (n+1)na \Rightarrow S_n = \frac{n(n+1)}{2} \cdot a. \tag{2.2}$$

(*ii*) *Isolating terms:* For a sum $S_n = \sum_{k=0}^{n} a_k$ compute S_{n+1} as $(\sum_{k=0}^{n} a_k) + a_{n+1}$ or $S_{n+1} = a_0 + \sum_{k=0}^{n} a_{n+1}$.

Example. (1) Let $S_n = \sum_{k=0}^{n} a^k$. Then, $S_{n+1} = S_n + a^{n+1} = 1 + \sum_{k=0}^{n} a^{k+1} = 1 + a \cdot S_n$, so we get $S_n = \frac{a^{n+1}-1}{a-1}$ if $a \neq 1$.

(2) Compute $S_n = \sum_{k=0}^{n} k \cdot 2^k$. Then, we get

$$\begin{aligned} S_{n+1} = S_n + (n+1)\cdot 2^{n+1} = 0 + \sum_{k=0}^{n}(k+1)\cdot 2^{k+1} = 2\overbrace{\sum_{k=0}^{n} k2^k}^{=S_n} + 2\overbrace{\sum_{k=0}^{n} 2^k}^{=2^{n+1}-1} \\ \Rightarrow S_n + (n+1)2^{n+1} = 2S_n + 2^{n+2} - 2 \\ \Leftrightarrow S_n = (n+1)2^{n+2} - 2^{n+1} + 2 \end{aligned} \tag{2.3}$$

(*iii*) *Induction:* If a formular for a sum is known, we can prove it by induction where we have seen several examples.

An non-trivial example for induction is the proof of the arithmetic-geometric inequality: For $a_1, \ldots, a_n \in \mathbb{R}_{\geq 0}$, we have $\sqrt[n]{a_1 \cdots a_n} \leq \frac{a_1 + \cdots + a_n}{n}$:

We prove the statement (P_n): For $a_1, \ldots, a_n \in \mathbb{R}_{\geq 0}$, we have $(a_1 \cdots a_n) \leq \left(\frac{a_1 + \cdots + a_n}{n}\right)^n$.
Proof by induction: (P_1): Nothing to show.
(P_2): Equivalent to $4a_1a_2 \leq a_1^2 + 2a_1a_2 + a_2^2 \iff 0 \leq (a_1 - a_2)^2$. We prove the statment by showing the following implications:

(*a*) $(P_n) \Rightarrow (P_{n-1})$,

(*b*) $((P_n)$ and $(P_2)) \Rightarrow (P_{2n})$.

(a): Consider

$$\prod_{k=1}^{n-1} a_k \cdot \sum_{k=1}^{n-1} \frac{a_k}{n-1} \overset{(P_n)}{\leq} \left(\frac{\sum_{k=1}^{n-1} a_k + \sum_{k=1}^{n-1} \frac{a_k}{n-1}}{n} \right)^n = \left(\frac{n \sum_{k=1}^{n-1} a_k}{n(n-1)} \right)^n = \left(\frac{1}{n-1} \right)^n \cdot \left(\sum_{k=1}^{n-1} a_k \right)^n, \tag{2.4}$$

so we get

$$\prod_{k=1}^{n-1} a_k \leq \left(\frac{1}{n-1} \right)^{n-1} \cdot \left(\sum_{k=1}^{n-1} a_k \right)^{n-1}. \tag{2.5}$$

(b): Let $a_1, \ldots, a_{2n} \in \mathbb{R}_{\geq 0}$. Then, we get

$$\begin{aligned} \prod_{k=1}^{2n} a_k &= \left(\prod_{k=1}^{n} a_k \right) \left(\prod_{k=n+1}^{2n} a_k \right) \overset{(P_n)}{\leq} \left(\sum_{k=1}^{n} \frac{a_k}{n} \right)^n \left(\sum_{k=n+1}^{2n} \frac{a_k}{n} \right)^n \\ &= \left(\sum_{k=1}^{n} \frac{a_k}{n} \sum_{k=n+1}^{2n} \frac{a_k}{n} \right) \overset{(P_2)}{\leq} \left(\frac{\sum_{k=1}^{n} \frac{a_k}{n} + \sum_{k=n+1}^{2n} \frac{a_k}{n}}{2} \right)^{2n} \\ &= \left(\frac{\sum_{k=1}^{2n} a_k}{2n} \right)^{2n} \end{aligned} \tag{2.6}$$

2.2 Difference and Sum operators

Sums can be considered as integrals of step functions. We want to develop a discrete analogon to the integral calculus. To do this, we first have to develop a discrete analogon to the differntial calculus.

Definition 2.1: (Translation) operators

For $a \in \mathbb{Z}$, the *translation operator* $E^a : \mathbb{R}^{\mathbb{Z}} \to \mathbb{R}^{\mathbb{Z}}$ mapp $f \in \mathbb{R}^{\mathbb{Z}}$ to $E^a(f)$ (or $E^a f$), where $E^a f(x) = f(x+a)$ for all $x \in \mathbb{Z}$. We write $I := E^0$ for the identity operator.
For two operators $P, Q : \mathbb{R}^{\mathbb{Z}} \to \mathbb{R}^{\mathbb{Z}}$, we define the *sum* by

$$(P+Q)f = Pf + Qf \tag{2.7}$$

and *multiplication by a scalar* $\alpha \in \mathbb{R}$ by setting $(\alpha P)f = \alpha \cdot Pf$.
The *compositon* QP is defined by $(QP)f = Q(Pf)$.

Definition 2.2: Δ, ∇:Difference operators

We define the *forward difference operator* $\Delta : \mathbb{R}^{\mathbb{Z}} \to \mathbb{R}^{\mathbb{Z}}$ as $\Delta := E^1 - I$, so for $f : \mathbb{Z} \to \mathbb{R}$, we have $\Delta f(x) = f(x+1) - f(x)$.
We also define the *backward difference operator* $\nabla : \mathbb{R}^{\mathbb{Z}} \to \mathbb{R}^{\mathbb{Z}}$ as $\nabla = I - E^1$, so for $f : \mathbb{Z} \to \mathbb{R}$, we have $\nabla f(x) = f(x) - f(x-1)$.

Example. (i) $\Delta x^{\underline{n}} = (x+1)^{\underline{n}} - x^{\underline{n}} = (x+1)x^{\underline{n-1}} - (x-n+1)x^{\underline{n-1}} = nx^{\underline{n-1}}$,
(ii) $\nabla x^{\overline{n}} = x^{\overline{n}} - (x-1)^{\overline{n}} = (x+n-1)x^{\overline{n-1}} - (x-1)x^{\overline{n-1}} = nx^{\overline{n-1}}$.

Definition 2.3: Falling/rising factorials for negative exponents

We generalize falling and rising factorials to negative exponents by setting (for $n \in \mathbb{N}\backslash\{0\}$):

$$x^{\underline{-n}} := \frac{1}{(x+1)(x+2)\cdots(x+n)}, \qquad x^{\overline{-n}} := \frac{1}{(x-1)(x-2)\cdots(x-n)}. \tag{2.8}$$

Theorem 2.4: Δ, ∇ for polynomials

For $n \in \mathbb{Z}$, we get

(*i*) $\Delta x^{\underline{n}} = n x^{\underline{n-1}}$,

(*ii*) $\nabla x^{\overline{n}} = n x^{\overline{n-1}}$.

Proof. For $n \geq 0$, we have seen this in the example above. So let now $n > 0$:

$$\Delta x^{\underline{-n}} = (x+1)^{\underline{-n}} - x^{\underline{-n}} = (x+1)x^{\underline{-n-1}} - (x+n-1)x^{\underline{-n-1}} = -n x^{\underline{-n-1}}, \tag{2.9}$$

and

$$\nabla x^{\overline{-n}} = x^{\overline{-n}} - (x-1)^{\overline{-n}} = (x-n-1)x^{\overline{-n-1}} - (x+1)x^{\overline{-n-1}} = -n x^{\overline{-n-1}}. \tag{2.10}$$

□

We will develop the difference calculus for the forward operator Δ.

Definition 2.5: $\sum$: Antiderivative; indefinite sum

For two functions $f, g : \mathbb{Z} \to \mathbb{R}$, we call f a *(discrete) antiderivative* of g if $\Delta f = g$. We write $f = \sum g$ and call f an *indefinite sum*

We get $\Delta f = g \iff f = \sum g$.

Theorem 2.6: „Main theorem of discrete differential- & integral calculus“

If f is an antiderivative of g, then for all $a \leq b \in \mathbb{Z}$, we have

$$\sum_{k=a}^{b} g(k) = f(b+1) - f(a). \tag{2.11}$$

Proof. Since $\Delta f = g$, we have $f(k+1) - f(k) = g(k)$ for $k \in \mathbb{Z}$. With that, we get

$$\sum_{k=a}^{b} g(k) = \sum_{k=0}^{n} (f(k+1) - f(k)) = f(b+1) - f(a). \tag{2.12}$$

□

The antiderivative is not unique (only up to additive constants).

Notation. For $f = \sum g$ and $a, b \in \mathbb{Z}$ with $a \leq b$, we write

$$\sum_{a}^{b} g(x) := f(x)\Big|_{x=a}^{b+1} := f(b+1) - f(a). \tag{2.13}$$

Thus, the notation is close to normal integrals, so we get

$$\sum_{a}^{b+1} g(x) = \sum_{k=a}^{b} g(k). \tag{2.14}$$

[22.10.2024, Lecture 5]
[24.10.2024, Lecture 6]

Observation. Both Δ and $\sum$ are linear operators, so $\Delta(\alpha f + \beta g) = \alpha\Delta f + \beta\Delta g$ and $\sum(\alpha f + \beta g) = \alpha \sum f + \beta \sum g$ for $f, g : \mathbb{Z} \to \mathbb{R}$ and $\alpha, \beta \in \mathbb{R}$.

Proposition 2.7. *For some $n \in \mathbb{Z}$, we have for $x \in \mathbb{N}_+$*

$$\sum x^{\underline{n}} = \begin{cases} \frac{x^{\underline{n+1}}}{n+1} & n \neq -1 \\ H_x & n = -1 \end{cases} \tag{2.15}$$

Proof. Since $\Delta x^{\underline{n+1}} = (n+1)x^{\underline{n}}$, we have $\sum x^{\underline{n}} = \frac{x^{\underline{n+1}}}{(n+1)}$ for $n \neq -1$.

A funciton f with $f = \sum x^{\underline{-1}}$ satisfies $\frac{1}{x+1} = x^{\underline{-1}} = \Delta f = f(x+1) - f(x)$, so we get that

$$H_x = \sum_{i=1}^{x} \frac{1}{i} \tag{2.16}$$

is an antiderivative of $x^{\underline{-1}}$. □

Application. Compute $\sum_{k=0}^{n} k^2$. We have $x^2 = x(x-1) + x = x^{\underline{2}} + x^{\underline{1}}$, hence, we get

$$\begin{aligned} \sum_{k=0}^{n} k^n = \sum_{0}^{n+1} x^2 = \sum_{0}^{n+1} x^{\underline{2}} + \sum_{0}^{n+1} x^{\underline{1}} &= \frac{x^{\underline{3}}}{3}\Big|_0 n + 1 + \frac{x^{\underline{2}}}{2}\Big|_0^{n+1} \\ &= \frac{(n+1)^{\underline{3}}}{3} + \frac{(n+1)^{\underline{2}}}{2} = \frac{n\left(n+\frac{1}{2}\right)(n+1)}{3}, \end{aligned} \tag{2.17}$$

or in more general,

$$\begin{aligned} x^m = \sum_{k=0}^{m} S_{m,k} x^{\underline{k}} \Rightarrow \sum_{k=0}^{n} k^m = \sum_{0}^{n+1} x^m &= \sum_{0}^{n+1} \left(S_{m,k} x^{\underline{k}}\right) \\ = \sum_{k=0}^{m} S_{m,k} \sum_{0}^{n+1} x^{\underline{k}} = \sum_{k=0}^{m} S_{m,k} \frac{x^{\underline{k+1}}}{k+1}\Big|_0^{n+1} \\ = \sum_{k=0}^{m} S_{m,k} \frac{(n+1)^{\underline{k+1}}}{k+1}. \end{aligned} \tag{2.18}$$

Example. (*i*) For $c \in \mathbb{R}$, we have

$$\Delta c^x = c^{x+1} - c^x = (c-1)c^x \Rightarrow \text{for } c \neq 1 : \sum c^k = \frac{c^x}{c-1}. \tag{2.19}$$

For $c = 2$, we have $\Delta 2^x = 2^x$ and $\sum 2^x = 2^x$.

(*ii*) For $x \in \mathbb{R}$ and $m \in \mathbb{Z}$, we have $\binom{x+1}{m+1} = \binom{x}{m} + \binom{x}{m+1}$ and hence, we get

$$\Delta\binom{x}{m+1} = \binom{x}{m} \quad \text{and} \quad \sum\binom{x}{m} = \binom{x}{m+1}. \tag{2.20}$$

Definition 2.8: Δ^n: n times applied Δ operator

For a function $f : \mathbb{Z} \to \mathbb{R}$, we define

$$\Delta^n f(x) := (E^1 - I)^n f(x) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} E^k f(x) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x+k). \quad (2.21)$$

In particular, for $x = 0$, we get

$$\Delta^n f(0) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(k). \quad (2.22)$$

Proposition 2.9 (Newton representation of polynomials)**.** *For a polynomial f of degree n, we have*

$$f(x) = \sum_{k=0}^{n} \frac{\Delta^k f(0)}{k!} x^{\underline{k}}. \quad (2.23)$$

Proof. Since the polynomials $x^{\underline{k}}$ are a basis of the vector space of polynomials, we can write f in a unique way as $f(x) = \sum_{k=0}^{n} b_k x^{\underline{k}}$. We have to show that $b_k = \frac{\Delta^k f(0)}{k!}$ for $k \in \{0, \dots, n\}$. We have $\Delta^k x^{\underline{i}} = i(i-1)(i-k+1)x^{\underline{i-k}} = i^{\underline{k}} x^{\underline{i-k}}$. We also get

$$\Delta^k f(x) = f(x) = \Delta^k \sum_{i=0}^{n} b_i x^{\underline{i}} = \sum_{i=0}^{n} b_i i^{\underline{k}} x^{\underline{i-k}}, \quad (2.24)$$

and for $x = 0$, we get

$$\Delta^k f(0) = \sum_{i=0}^{n} b_i i^{\underline{k}} 0^{\underline{i-k}} = b_k k^{\underline{k}} = b_k k!, \quad (2.25)$$

because $i^{\underline{k}} = 0$ for $k > i$ and $0^{\underline{i-k}} = 0$ for $k < i$. Thus, we get $b_k = \frac{\Delta^k f(0)}{k!}$.[1] □

Corollary 2.10. *For $n, k \in \mathbb{N}$, we get*

$$S_{n,k} = \frac{1}{n!} \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} i^n. \quad (2.26)$$

Proof. We have

$$x^n = \sum_{k=0}^{n} S_{n,k} x^{\underline{k}}, \quad (2.27)$$

so by Proposition 2.9 applied to $f(x) = x^n$, we get

$$S_{n,k} = \frac{\Delta^k f(0)}{k!} = \frac{1}{k!} \sum_{i=0}^{k} (-1)^{k-i} (-1)^{k-i} \binom{k}{i} i^n. \quad (2.28)$$

□

Theorem 2.11: Partial summation

For functions $u, v : \mathbb{Z} \to \mathbb{R}$, we have

$$\sum (u \cdot \Delta v) = u \cdot v - \sum (\Delta u \cdot v). \quad (2.29)$$

[1] Student: „Shouldn't it be $k > i$ and $k < i$?" – Brenner: „Yes, indeed. Don't trust on what I'm writing today. *thinking* Maybe not only today." – Ulrich Brenner, 24.10.2024

Proof. We have

$$\begin{aligned}\Delta(u\cdot v)(x) &= u(x+1)v(x+1)-u(x)v(x)\\ &= u(x)(v(x+1)-v(x))+v(x+1)(u(x+1)-u(x))\\ &= u(x)\Delta v(x)+Ev(x)\cdot\Delta u(x)\\ \Rightarrow \Delta(uv) &= u\Delta v+(Ev)\Delta u.\end{aligned} \tag{2.30}$$

Now apply $\sum$ on this equation. □

Applications.

(*i*) Compute $\sum_{k=0}^{n} k\cdot 2^k$: We want $\sum x2^x$. Apply Theorem 2.11 with $u(x)=x$ and $v(x)=2^x$, so $\Delta u(x)=1$ and $\Delta v(x)=2^x$. With that, we get

$$\begin{aligned}\sum_{k=0}^{n} k\cdot 2^k &= \sum_{0}^{n+1} x\cdot 2^x = x2^x\Big|_0^{n+1} - \sum_{0}^{n+1} 2^{x+1} = x2^x\Big|_0^{n+1} - 2\cdot 2^x\Big|_0^{n+1}\\ &= (n+1)2^{n+1} - 2\cdot 2^{n+1} + 2 = (n-1)2^{n+1}+2.\end{aligned} \tag{2.31}$$

(*ii*) Compute $\sum_{k=0}^{n} H_k$: Set $u(x)=H_x$ and $\Delta v(x)=1=x^{\underline{0}}$, so $\Delta u(x)=\frac{1}{1+x}$ and $v(x)=x$. Then, we get using Theorem 2.11,

$$\begin{aligned}\sum_{k=1}^{n} H_k &= \sum_{1}^{n+1} H_x x^{\underline{0}} = H_x x\Big|_0^{n+1} - \sum_{1}^{n+1}(x+1)\cdot\frac{1}{x+1}\\ &= H_{n+1}\cdot(n+1) - 1 - x\Big|_1^{n+1} = H_{n+1}(n+1) - 1 - (n+1) + 1\\ &= (n+1)\cdot(H_{n+1}-1).\end{aligned} \tag{2.32}$$

(*iii*) Compute $\sum_{k=1}^{n}\binom{k}{m}H_k$ for some $m\in\mathbb{N}\setminus\{0\}$: Choose $u(x)=H_x$ and $\Delta v(x)=\binom{x}{m}$. Then, $\Delta u(x)=\frac{1}{x+1}$ and $v(x)=\binom{x}{m+1}$. Apply Theorem 2.11 and we get

$$\begin{aligned}\sum_{k=1}^{n}\binom{k}{m}H_k &= \sum_{1}^{n+1}\binom{x}{m}H_x = H_x\binom{x}{m+1}\Big|_0^{n+1} - \sum_{1}^{n+1}\binom{x+1}{m+1}\cdot\frac{1}{x+1}\\ &= H_{n+1}\binom{n+1}{m+1} - H_1\binom{1}{m+1} - \frac{1}{m+1}\sum_{1}^{n+1}\binom{x}{m}\\ &= H_{n+1}\binom{n+1}{m+1} - \frac{1}{m+1}\binom{x}{m+1}\Big|_1^{n+1}\\ &= H_{n+1}\binom{n+1}{m+1} - \frac{1}{m+1}\left(\binom{n+1}{m+1} - \binom{1}{m+1}\right)\\ &= \binom{n+1}{m+1}\left(H_{n+1} - \frac{1}{n+1}\right).\end{aligned} \tag{2.33}$$

2.3 Inversions

Definition 2.12: Basis sequence

A *basis sequence* is a sequence of polynomials $(p_i)_{i\in\mathbb{N}}$ where p_i is a polynomial of degree i.

Examples are $(x^n)_{n\in\mathbb{N}}, (x^{\overline{n}})_{n\in\mathbb{N}}, (x^{\underline{n}})_{n\in\mathbb{N}}$.

When $(p_i)_{i\in\mathbb{N}}$ is a basis seqnence and p is a polynomial of degree n, then there are unique numbers $a_0, \dots, a_n$ such that

$$p(x) = \sum_{k=0}^{n} a_k p_k(x). \tag{2.34}$$

If $(p_n)_{n\in\mathbb{N}}$ and $(q_n)_{n\in\mathbb{N}}$ are basis sequences, then there are unique numbers $a_{n,k}$ and $b_{n,k}$ for $k = 0, \dots, n$ with

$$q_n(x) = \sum_{k=0}^{n} a_{n,k} p_k(x) \quad \text{and} \quad p_n(x) = \sum_{k=0}^{n} b_{n,k} q_k(x). \tag{2.35}$$

Definition 2.13: Connection coefficients

The numbers $a_{n,k}$ and $b_{n,k}$ are called *conneciton coefficients*.

[24.10.2024, Lecture 6]
[29.10.2024, Lecture 7]

Convention. For $n < k$, we set $a_{n,k} = b_{n,k} = 0$. With that, we get

$$q_n(x) = \sum_{k=0}^{n} a_{n,k} p_k(x) = \sum_{k=0}^{n} a_{n,k} \sum_{m=0}^{k} b_{k,m} q_m(x) = \sum_{m=0}^{n} q_m(x) \sum_{k=m}^{n} a_{n,k} b_{k,m}. \tag{2.36}$$

With that, we get that

$$\sum_{k=m}^{n} a_{n,k} b_{k,m} = \sum_{k=1}^{m} a_{m,k} b_{k,m} = \delta_{m,n} = \begin{cases} 1 & m = n \\ 0 & \text{else.} \end{cases} \tag{2.37}$$

We obsereve: For $A_n = (a_{ij}), B_n = (b_{ij}) \in \mathbb{R}^{n\times n}$, we get that $A_n B_n = I_n$, so they are inverse to eachother.

Theorem 2.14: Basis sequences from connection coefficients

Let $(p_n)_{n\in\mathbb{N}}, (q_n)_{n\in\mathbb{N}}$ be basis sequences with coefficients $a_{n,k}$ and $b_{n,k}$. Then, for sequences $(u_n)_{n\in\mathbb{N}}, (v_n)_{n\in\mathbb{N}}$, the following statements are equivalent:

(i) For all $n \in \mathbb{N}$, we get $v_n = \sum_{k=0}^{n} a_{n,k} u_k$,

(ii) For all $n \in \mathbb{N}$, we get $u_n = \sum_{k=0}^{n} b_{n,k} v_k$.

Proof. Let $n \in \mathbb{N}$. The matrices A_n and B_n are inverse to eachother, so for any vectors $u = (u_1, \dots, u_n)$ and $v = (v_1, \dots, v_n)$, we have $v = A_n u \iff u = B_n v$. □

Example.

(i) *Stirling numbers*: For the basis sequences $(x^n)_{n\in\mathbb{N}}$ and $(x^{\underline{n}})_{n\in\mathbb{N}}$, we know already that $x^n = \sum_{k=0}^{n} S_{n,k} x^{\underline{k}}$ and $x^{\underline{n}} = \sum_{k=0}^{n} (-1)^{n-1} s_{n,k} x^k$. This means, that $S_{n,k}$ and $(-1)^{n-k} s_{n,k}$ are the connection coefficients of the basis sequences $(x^n)_{n\in\mathbb{N}}$ and $(x^{\underline{n}})_{n\in\mathbb{N}}$. With this, we get

$$\sum_{k=1}^{n} S_{n,k} (-1)^{k-m} s_{k,m} = \begin{cases} 1 & m = n \\ 0 & m \neq n. \end{cases} \tag{2.38}$$

Moreover, by Theorem 2.14 we get that for any sequences $(u_n)_{n\in\mathbb{N}}, (v_n)_{n\in\mathbb{N}}$, we have that

$$\left(\forall n \in \mathbb{N} : v_n = \sum_{k=0}^{n} S_{n,k} u_k\right) \iff \left(\forall n \in \mathbb{N} : u_n = \sum_{k=0}^{n} (-1)^{n-k} s_{m,k} v_k\right) \tag{2.39}$$

(*ii*) *Binomial coefficients:* By the binomial theorem, for $n \in \mathbb{N}$, we have

$$x^n = ((x-1)+1)^n = \sum_{k=0}^{n} \binom{n}{k} (x-1)^k \tag{2.40}$$

and

$$(x-1)^n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} x^k. \tag{2.41}$$

Again, by Theorem 2.14, we get that $\binom{n}{k}$ and $(-1)^{n-k}\binom{n}{k}$ are connection coefficients of the basis sequences $((x-1)^n)_{n\in\mathbb{N}}$ and $(x^n)_{n\in\mathbb{N}}$. This gives us

$$\sum_{k\geq 0} \binom{n}{k} (-1)^{k-m} \binom{k}{m} = \begin{cases} 1 & n = m \\ 0 & n \neq m, \end{cases} \tag{2.42}$$

so for any sequences $(u_n)_{n\in\mathbb{N}}, (v_n)_{n\in\mathbb{N}}$, we have that

$$\left(\forall n \in \mathbb{N} : v_n = \sum_{k=0}^{n} \binom{n}{k} u_k \right) \iff \left(\forall n \in \mathbb{N} : u_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} v_k \right). \tag{2.43}$$

This equivalence is called *binomial inversion*.

By replacing u_n by $(-1)^n u_n$, we get a more symmetric version: For sequences $(u_n)_{n\in\mathbb{N}}$ and $(v_n)_{n\in\mathbb{N}}$, we have

$$\left(\forall n \in \mathbb{N} : v_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} u_k \right) \iff \left(\forall n \in \mathbb{N} : u_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} v_k \right). \tag{2.44}$$

Application. Consider the derangement numbers. We have

$$n! = \sum_{k=0}^{n} D_k \tag{2.45}$$

(because we have $\binom{n}{n-k} D_k = \binom{n}{k} D_k$) ways to permute $\{1, \ldots, n\}$ with exactly $n-k$ fixed points. This means, that by binomial inversion (2.43), we get

$$D_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} k! = n! \cdot \sum_{k=0}^{n} \frac{(-1)^{n-k}}{(n-k)!} = n! \cdot \sum_{k=0}^{n} \frac{(-1)^k}{k!}. \tag{2.46}$$

Chapter 3

Recursions

3.1 General recursion formula

Theorem 3.1: General recursion formula

Let $T_0, T_1, T_2, \dots$ bea sequence that is given by numbers β and a_n, b_n with $a_n \neq 0$ (with $n \in \mathbb{N} \setminus \{0\}$) and the following recursions:

- $T_0 = \beta$,
- $T_n = a_n T_{n-1} + b_n$ *for* $n \geq 1$.

Then, for all $n \in \mathbb{N}$, we can write T_n directly by

$$T_n = \prod_{k=1}^{n} a_k \left(\sum_{k=1}^{n} \frac{b_k}{\prod_{i=1}^{k} a_i} + \beta \right). \tag{3.1}$$

Proof. Induction in n. If $n = 0$, it follows from $T_0 = \beta$ and rules for empty sums and products. If $n \geq 1$, we get

$$T_n = a_1 T_{n-1} + b_n = a_n \prod_{k=1}^{n-1} a_k \left(\sum_{k=1}^{n} \frac{b_k}{\prod_{i=1}^{k} a_i} + T_0 \right) + b_n = \prod_{k=1}^{n} a_k \left(\sum_{k=1}^{n} \frac{b_k}{\prod_{i=1}^{k} a_i} + T_0 \right). \tag{3.2}$$

□

Corollary 3.2. *Let $T_0, T_1, \dots$ be a sequence that is given by numbers a, b, β with $a \neq 1$ and the following recursion:*

- $T_0 = \beta$,
- $T_n = aT_{n-1} + b$ *for* $n \geq 1$.

Then, for all $n \in \mathbb{N}$, we get that

$$T_n = a^n T_0 + b \frac{a^{n-1} - 1}{a - 1}. \tag{3.3}$$

Proof. If $a = 0$, we have $T_0 = \beta$ and $T_n = b$ for $n \geq 1$, so the statement holds.

If $a \neq 0$, and the statment follows from Theorem 3.1 and the fact that

$$\sum_{k=1}^{n} \frac{1}{a^k} = -1 + \sum_{k=0}^{n} \frac{1}{a^k} = -1 + \frac{1 - \left(\frac{1}{a}\right)^{n+1}}{1 - a} = \frac{1 - \left(\frac{1}{a}\right)^{n}}{a - 1}, \tag{3.4}$$

so

$$a^n b \sum_{k=1}^{n} \frac{1}{a^k} = b \frac{a^{n-1}}{a - 1}. \tag{3.5}$$

□

Application. Consider the derangement numbers D_n. We have

$$D_n = (n-1)(D_{n-1} + D_{n-2}) \qquad \text{for } n \geq 1, \text{ with } D_0 := 1. \tag{3.6}$$

To see this, note that the set of fixed-point free n-permutations can be decomposed into

- $(n-1)D_{n-1}$ permutations where 1 is contained in a cycle of length 2,
- $(n-1)D_{n-2}$ permutations where 1 is contained in a cycle of length ≥ 3.

With this, we get that $D_n - nD_{n-1} = -(D_{n-1} - (n-1)D_{n-2}) = D_{n-2} - (n-2)D_{n-3}$ and so on, until we get $D_n - nD_{n-1} = (-1)^{n-1}(D_1 - D_0) = (-1)^n$, so we get

$$D_n = nD_{n-1} + (-1)^n. \tag{3.7}$$

We can now apply Theorem 3.1 to $\beta = 1, a_n = n$ and $b_n = (-1)^n$ and recieve

$$D_n = \prod_{k=1}^{n} a_k \left(\sum_{k=1}^{n} \frac{b_k}{\prod_{i=1}^{k} a_i} + \beta \right) = n! \left(\sum_{k=1}^{n} \frac{(-1)^k}{k!} + 1 \right) = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}. \tag{3.8}$$

3.2 Generating functions

Definition 3.3: Generating function

A *generating function* of a sequence $(a_n)_{n\in\mathbb{N}}$ is the formal expression

$$\sum_{n\geq 0} a_n z^n. \tag{3.9}$$

Remark. *Formal expression* means in particular that we don't care where the sum converges.

3.2.1 Calculating with generating functions

Example.

(*i*) The sum of two generation functions $\sum_{n\geq 0} a_n z^n$ and $\sum_{n\geq 0} b_n z^n$ is $\sum_{n\geq 0}(a_n + b_n)z^n$.

(*ii*) The product of two generation functions $\sum_{n\geq 0} a_n z^n$ and $\sum_{n\geq 0} b_n z^n$ is given by the convolution of the sequences $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$, so we get

$$\left(\sum_{n\geq 0} a_n z^n \right) \left(\sum_{n\geq 0} b_n z^n \right) = \sum_{n\geq 0} \left(\sum_{k=0}^{n} a_k b_{n-k} \right) z^n \tag{3.10}$$

(*iii*) $\sum_{n\geq 0} a_n z^n = 0$ is the additive identity and $\sum_{n\geq 0} a_n z^n = 1$ is the multiplicative identity.

[29.10.2024, Lecture 7]
[31.10.2024, Lecture 8]

Appendix

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